Major brain disorders - Insight 46 Phase 1

Sarah Keuss | 15/09/2021

Document details

Categories of variables	Major brain disorder
Summary of work undertaken	Major brain disorders were defined as described below. Data was cleaned and summarised as per output variables below.
Names of source variables	General:, Subject, Date, Cognitive, and, neurological, history:, diagnosis1, other1, diagnosis2, other2, Radiology, reads:, infarction, haemorrhage, lesion, aneurysm, colloidCyst, hydrocephalus, sinusDisease, other, comments, flaggedForReview
Output variables	mbraindis_i46p1, diag1_brdis_i46p1, diag2_brdis_i46p1

Overview

<u>For Submission to the NSHD Scientific Support Team</u>

Summary

Major brain disorders were determined by a consensus panel including consultant neurologists and study clinicians who had reviewed the participants. Decisions were made based on self-reported diagnosis **(diagnosis1, other1, diagnosis2, other2)**, radiological reads **(infarction, haemorrhage, lesion, aneurysm, colloidCyst, hydrocephalus, sinusDisease, other, comments, flaggedForReview)** and clinician opinion.

Participants were classified as having a major brain disorder if they had a clinical or radiological diagnosis of a significant neurological or psychiatric condition. These included but were not limited to:

1. Neurodegenerative disorders
2. Multiple sclerosis
3. Epilepsy requiring active treatment
4. Psychiatric disorder requiring hospital admission, anti-psychotic medication, or electroconvulsive therapy
5. Major neurosurgery
6. Clinical or radiological (cortical) stroke

List any papers in which cleaned/derived variables have been used

Parker T, Cash DM, Lane C, Lu K, Malone IB, Nicholas JM, James S, Keshavan A, Murray-Smith H, Wong A, Buchannan S, Keuss S, Sudre CH, Thomas D, Crutch S, Bamiou DE, Warren JD, Fox NC, Richards M, Schott JM. Pure tone audiometry and cerebral pathology in healthy older adults. J Neurol Neurosurg Psychiatry. 2020 Feb;91(2):172-176. doi: 10.1136/jnnp-2019-321897. Epub 2019 Nov 7. PMID: 31699832; PMCID: PMC6996095.

Parker TD, Cash DM, Lane CA, Lu K, Malone IB, Nicholas JM, James SN, Keshavan A, Murray-Smith H, Wong A, Buchanan SM, Keuss SE, Sudre CH, Thomas DL, Crutch SJ, Fox NC, Richards M, Schott JM. Amyloid β influences the relationship between cortical thickness and vascular load. Alzheimers Dement (Amst). 2020 Apr 17;12(1):e12022. doi: 10.1002/dad2.12022. PMID: 32313829; PMCID: PMC7163924.

Lu K, Nicholas JM, James SN, Lane CA, Parker TD, Keshavan A, Keuss SE, Buchanan SM, Murray-Smith H, Cash DM, Sudre CH, Malone IB, Coath W, Wong A, Henley SMD, Fox NC, Richards M, Schott JM, Crutch SJ. Increased variability in reaction time is associated with amyloid beta pathology at age 70. Alzheimers Dement (Amst). 2020 Aug 10;12(1):e12076. doi: 10.1002/dad2.12076. PMID: 32789161; PMCID: PMC7416668.

Lane CA, Barnes J, Nicholas JM, Sudre CH, Cash DM, Malone IB, Parker TD, Keshavan A, Buchanan SM, Keuss SE, James SN, Lu K, Murray-Smith H, Wong A, Gordon E, Coath W, Modat M, Thomas D, Richards M, Fox NC, Schott JM. Associations Between Vascular Risk Across Adulthood and Brain Pathology in Late Life: Evidence From a British Birth Cohort. JAMA Neurol. 2020 Feb 1;77(2):175-183. doi: 10.1001/jamaneurol.2019.3774. PMID: 31682678; PMCID: PMC6830432.

Lu K, Nicholas JM, Collins JD, James SN, Parker TD, Lane CA, Keshavan A, Keuss SE, Buchanan SM, Murray-Smith H, Cash DM, Sudre CH, Malone IB, Coath W, Wong A, Henley SMD, Crutch SJ, Fox NC, Richards M, Schott JM. Cognition at age 70: Life course predictors and associations with brain pathologies. Neurology. 2019 Dec 3;93(23):e2144-e2156. doi: 10.1212/WNL.0000000000008534. Epub 2019 Oct 30. PMID: 31666352; PMCID: PMC6937487.

Parker TD, Cash DM, Lane CAS, Lu K, Malone IB, Nicholas JM, James SN, Keshavan A, Murray-Smith H, Wong A, Buchanan SM, Keuss SE, Sudre CH, Modat M, Thomas DL, Crutch SJ, Richards M, Fox NC, Schott JM. Hippocampal subfield volumes and pre-clinical Alzheimer's disease in 408 cognitively normal adults born in 1946. PLoS One. 2019 Oct 17;14(10):e0224030. doi: 10.1371/journal.pone.0224030. PMID: 31622410; PMCID: PMC6797197.

Keshavan A, Pannee J, Karikari TK, Rodriguez JL, Ashton NJ, Nicholas JM, Cash DM, Coath W, Lane CA, Parker TD, Lu K, Buchanan SM, Keuss SE, James SN, Murray-Smith H, Wong A, Barnes A, Dickson JC, Heslegrave A, Portelius E, Richards M, Fox NC, Zetterberg H, Blennow K, Schott JM. Population-based blood screening for preclinical Alzheimer's disease in a British birth cohort at age 70. Brain. 2021 Mar 3;144(2):434-449. doi: 10.1093/brain/awaa403. PMID: 33479777; PMCID: PMC7940173.

James SN, Nicholas JM, Lane CA, Parker TD, Lu K, Keshavan A, Buchanan SM, Keuss SE, Murray-Smith H, Wong A, Cash DM, Malone IB, Barnes J, Sudre CH, Coath W, Prosser L, Ourselin S, Modat M, Thomas DL, Cardoso J, Heslegrave A, Zetterberg H, Crutch SJ, Schott JM, Richards M, Fox NC. A population-based study of head injury, cognitive function and pathological markers. Ann Clin Transl Neurol. 2021 Apr;8(4):842-856. doi: 10.1002/acn3.51331. Epub 2021 Mar 11. PMID: 33694298; PMCID: PMC8045921.

Lane CA, Barnes J, Nicholas JM, Baker JW, Sudre CH, Cash DM, Parker TD, Malone IB, Lu K, James SN, Keshavan A, Buchanan S, Keuss S, Murray-Smith H, Wong A, Gordon E, Coath W, Modat M, Thomas D, Hardy R, Richards M, Fox NC, Schott JM. Investigating the relationship between BMI across adulthood and late life brain pathologies. Alzheimers Res Ther. 2021 Apr 30;13(1):91. doi: 10.1186/s13195-021-00830-7. PMID: 33941254; PMCID: PMC8091727.

Lu K, Nicholas JM, Weston PSJ, Stout JC, O'Regan AM, James SN, Buchanan SM, Lane CA, Parker TD, Keuss SE, Keshavan A, Murray-Smith H, Cash DM, Sudre CH, Malone IB, Coath W, Wong A, Richards M, Henley SMD, Fox NC, Schott JM, Crutch SJ. Visuomotor integration deficits are common to familial and sporadic preclinical Alzheimer's disease. Brain Commun. 2021 Jan 25;3(1):fcab003. doi: 10.1093/braincomms/fcab003. PMID: 33615219; PMCID: PMC7882207.

Pavisic IM, Lu K, Keuss SE, James SN, Lane CA, Parker TD, Keshavan A, Buchanan SM, Murray-Smith H, Cash DM, Coath W, Wong A, Fox NC, Crutch SJ, Richards M, Schott JM. Subjective cognitive complaints at age 70: associations with amyloid and mental health. J Neurol Neurosurg Psychiatry. 2021 May 25:jnnp-2020-325620. doi: 10.1136/jnnp-2020-325620. Epub ahead of print. PMID: 34035132.

Lane CA, Barnes J, Nicholas JM, Sudre CH, Cash DM, Parker TD, Malone IB, Lu K, James SN, Keshavan A, Murray-Smith H, Wong A, Buchanan SM, Keuss SE, Gordon E, Coath W, Barnes A, Dickson J, Modat M, Thomas D, Crutch SJ, Hardy R, Richards M, Fox NC, Schott JM. Associations between blood pressure across adulthood and late-life brain structure and pathology in the neuroscience substudy of the 1946 British birth cohort (Insight 46): an epidemiological study. Lancet Neurol. 2019 Oct;18(10):942-952. doi: 10.1016/S1474-4422(19)30228-5. Epub 2019 Aug 20. Erratum in: Lancet Neurol. 2020 Jul;19(7):e6. PMID: 31444142; PMCID: PMC6744368.

Variables in this document

3 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Health and medical history [6040] › Personal history of neurological disorder [60401]		
diag1_brdis_i46p1	Specifies diagnosis if yes to major brain disorder - Insight46 Phase 1	topsheet field
diag2_brdis_i46p1	Specifies second diagnosis if more than one major brain disorder - Insight46 phase 1	topsheet field
mbraindis_i46p1	Classified as having a major brain disorder - Insight46 Phase 1	topsheet field

2 medium-confidence matches

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Heart and circulatory conditions [51014]		
stroke	Have you had a stroke in the last 10 years - at age 53 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan